

APPLE NEURAL

The private AI computer for the Apple ecosystem

Independent product concept • October 2026



Apple Neural

The private AI computer for your entire home.

- Private by design**
Your data stays in your home.
- Shared for your family**
Multiple users, personalized experiences.
- More capable AI models**
Larger models, faster responses.
- Works across all Apple devices**
iPhone, iPad, Mac, Apple Watch, Apple TV, HomePod and more.

Apple Neural
Quiet. Powerful. Always there.

Models & Specifications

Choose the right level of intelligence for your home.

	Apple Neural	Apple Neural Pro	Apple Neural Max
Price	\$499	\$799	\$1,299
Use Case	Great for everyday AI in your home.	More power for advanced models.	Maximum capability for the most demanding AI experiences.
Chip	Apple S1	Apple S1 Pro	Apple S1 Max
CPU / GPU	8-core CPU / 10-core GPU	12-core CPU / 16-core GPU	16-core CPU / 32-core GPU
Neural Engine	16-core	32-core	64-core
Unified Memory	32 GB	64 GB	128 GB
Storage	512 GB	1 TB	2 TB
Connectivity	Wi-Fi 7, Ethernet	Wi-Fi 7, Ethernet	Wi-Fi 7, Ethernet
Power Consumption	~15W (idle)	~30W (idle)	~30W (idle)
Size	7.7 x 7.7 x 2.0 in	7.7 x 7.7 x 2.0 in	7.7 x 7.7 x 2.0 in

How it works

Your devices connect to Apple Neural for advanced AI.



Devices: iPhone, iPad, Mac, Apple Watch, Apple TV, HomePod

Apple Neural: Runs Apple's AI models locally in your home.

Connectivity: Your Home Network (Wi-Fi or Ethernet)

Access from anywhere: Encrypted connection over the internet.

Real use cases. A smarter home.

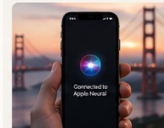
- Personal productivity**
Summarize emails, write, research, plan your day.
"Draft a reply and schedule it."
- Family assistance**
Personalized experiences for each family member.
"Help me with my homework."
- Home automation**
Understand and control your HomeKit devices.
"Turn on the living room lights and set the temperature to 72°."
- Creativity and learning**
Generate, create and explore new ideas.
"Create a 3D model of a mountain cabin."
- Entertainment**
Find, summarize and recommend content you'll love.
"Find a good movie for family night."

Privacy by design

- Local inference**
Your data stays in your home.
- Per-user isolation**
Each family member has a private space.
- End-to-end encrypted access**
Secure connection when you're away.
- No data sent to Apple**
Your personal data never leaves your home.

Works even when you're away

Securely connect to your Apple Neural over the internet.



Thoughtfully designed

Blends beautifully into any space.



Whisper quiet • Energy efficient • Minimal design

THE IDEA

As AI models become too capable to run comfortably on every personal device, Apple could add a new layer between device and cloud: a dedicated, always-on Apple Silicon compute hub that runs larger models locally for the entire household.

Your devices. Your home. One Neural.

1. Executive Summary

Apple's next AI challenge is not simply model quality. It is where those models should run. Phones, watches and other edge devices are constrained by battery, thermals, memory and physical size, while sending everything to the cloud creates latency, connectivity and privacy trade-offs.

Apple Neural proposes a small, purpose-built Apple Silicon appliance that sits inside the home and provides shared local AI inference to iPhone, iPad, Mac, Apple Watch, Apple TV and HomePod. It would be closer to a Mac mini in physical form, but its user experience would be closer to a HomePod: plug it in, connect it to the home, and let the ecosystem use it automatically.

THE CORE THESIS

Apple should not force every device to become an AI supercomputer. Give the household one powerful, private AI computer—and let every Apple device use it.

2. The Problem

AI capability is becoming a hardware problem

- Larger models require substantially more memory and compute than an iPhone or Apple Watch can economically dedicate to inference.
- Increasing on-device capability raises cost, power consumption, thermal constraints and device complexity.
- Cloud inference provides scale, but introduces network latency and requires sensitive prompts or context to leave the home.
- A household may already own several capable Apple devices, but their compute resources are fragmented rather than intentionally pooled.

A missing layer in the stack

Today: Device → Cloud

Proposed: Device → Apple Neural → Cloud (only when needed)

3. The Product

Apple Neural is a compact Apple-designed compute appliance built specifically for AI inference. It would have no screen and little direct interaction. Its job is to quietly make every Apple device in the home more capable.

- Apple Silicon optimized for inference, with a large Neural Engine/GPU and high-bandwidth unified memory.

- 32 / 64 / 128 GB memory configurations to support progressively larger models and more simultaneous users.
- Fast SSD for model storage, local indexes and optional household AI data.
- Wi-Fi 7 and Ethernet for high-throughput, low-latency local communication.
- Low idle power and quiet thermal design for always-on operation.
- Automatic device discovery and authentication through the Apple ecosystem.
- No requirement for users to understand model hosting, networking or AI infrastructure.

Apple Neural
The private AI computer for your entire home.
More capable Siri. Greater privacy. Smarter experiences. Together.

Private by design | Shared across your home | Powerful Apple silicon | Works with all Apple devices

Models & Specifications
Choose the right level of intelligence for your home.

	Apple Neural \$499 Powerful. Essential.	Apple Neural Pro \$799 More capable. For growing needs.	Apple Neural Max \$1,299 Maximum intelligence. For power users.
Apple Silicon	Apple N1	Apple N1 Pro	Apple N1 Max
Neural Engine / GPU	16-core	32-core	64-core
Unified Memory	32 GB	64 GB	128 GB
Storage	512 GB	1 TB	2 TB
Connectivity	Wi-Fi 7, Ethernet	Wi-Fi 7, Ethernet	Wi-Fi 7, Ethernet
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One Neural. Every device. A more capable Siri.
Apple Neural runs advanced AI models locally in your home, powering all your Apple devices with private, powerful intelligence.

iPhone iPad Mac Apple Watch Apple TV HomePod

How Apple Neural works
Your devices connect to Apple Neural for advanced AI, at home or anywhere.

Local network (Wi-Fi / Ethernet) | Secure remote access (when you're away)

Intelligence for the whole family.
Each family member gets a private and personalized experience, with shared compute for a smarter home.

Vedant: Work, productivity, coding help
Sarah: Personal planning, health & wellness
Aarav: Homework help, learning
Dad: News, research, everyday questions

A 3-layer intelligence architecture
The best experience, always. On-device when possible, Apple Neural when needed, and Private Cloud for the most complex tasks.

On-device AI | Apple Neural | Apple Private Cloud Compute

Privacy by design
Your data stays in your home, in your control.

- Local inference: Your data never leaves your home (unless you choose).
- Per-user isolation: Each family member has a separate, private space.
- Encrypted remote access: Secure connection when you're away.
- No data sold. No training on your personal data. Your data is yours.

Real-world use cases
More capable Siri. A smarter, more helpful home.

Summarize emails and documents | Plan trips and manage schedules | Homework help for kids | Health & fitness guidance | Smart home control (HomeKit) | Analyze photos and videos | Coding and creative projects | Personalized recommendations

Illustrative concept specifications and use cases; not actual Apple product specifications.

4. How It Works

Local network inference

The inference request would normally travel over the home's local Wi-Fi or Ethernet network—not Bluetooth. Bluetooth could assist with discovery or proximity, but sustained model traffic belongs on the local network.

1. The user asks Siri or invokes an AI feature on an Apple device.
2. The operating system classifies the request by complexity, sensitivity, latency requirements and available local compute.
3. Simple tasks run directly on the device.
4. Larger tasks are securely routed to Apple Neural over the local network.

Apple Neural — Independent product concept

- 5. The Apple Neural device runs the appropriate Apple model and returns the result.
- 6. If the task exceeds local capability, the system can escalate to Apple's private cloud infrastructure.

Three-tier inference

Tier	Where	Best for	Priority
1	Apple device	Fast, lightweight tasks	Lowest latency
2	Apple Neural	Large/private models	Privacy + capability
3	Apple private cloud	Largest / specialized workloads	Maximum capability

5. One Compute Layer for the Entire Family

The strongest consumer proposition may be the household rather than the individual. One Apple Neural device can serve multiple Apple accounts while maintaining strict per-user isolation.

- Family Sharing becomes a natural identity and entitlement layer.
- Each person receives a separate AI context, permissions boundary and personalization profile.
- Parents can manage child access and age-appropriate capabilities without exposing adult private context.
- Compute is shared; personal data and conversational context remain logically separated.

THE CONSUMER MAGIC

Buy one box. Every Apple device in the home becomes smarter.

6. What People Could Do With It

A more capable Siri — Ask complex questions, reason over documents, summarize long material, plan trips, compare options and perform multi-step tasks without requiring every device to host the largest model.

Personal AI — The home can securely provide a larger model with context from the user's permitted devices, files and preferences.

Smarter Home — Apple Neural can act as the reasoning layer for HomeKit: understanding natural-language requests, preferences and multi-step automations.

Family AI — Each family member gets a personalized experience while sharing the same physical compute resource.

Mac companion — A MacBook can offload selected inference workloads when it wants more memory or sustained compute without draining battery.

Apple Watch — The Watch can delegate complex reasoning rather than attempting to run every advanced model locally.

Always-available agents — Future Apple agents could execute longer-running local workflows while the user is away from the device that initiated them.

7. It Still Works When You're Away

When the user leaves home, the same architecture can remain available through an authenticated, end-to-end encrypted connection. The request can travel over the internet to the user's Apple Neural device, the model runs there, and the response returns to the device.

IMPORTANT

Remote access should be opt-in, strongly authenticated and privacy-preserving. The concept does not require exposing the home device directly to the public internet.

8. Privacy by Design

- Local inference: prompts and model inputs can remain inside the home for eligible tasks.
- Per-user isolation: separate identities, permissions, contexts and encrypted storage.
- Encrypted transport: device-to-hub and remote connections are authenticated and encrypted.
- Minimal telemetry: Apple can design the product so personal inference data is not used for model training.
- User control: users can see which device handled a request and whether it stayed local or escalated to Apple's private cloud.

9. Product Line & Positioning

Illustrative positioning—not a prediction of Apple's actual specifications or pricing.

Model	Memory	Positioning	Illustrative price	Primary user
Apple Neural	32 GB	Entry household AI	\$499	Most homes
Apple Neural Pro	64 GB	Power users / families	\$799	Heavy AI use

Apple Neural Max	128 GB	Large models / advanced agents	\$1,299	Enthusiasts / pros
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Why Apple could make this work

- Apple controls the silicon, operating systems, identity layer, AI frameworks, devices and home ecosystem.
- The company can optimize the entire inference stack rather than relying on third-party accelerators.
- The same compute can increase the value of multiple existing products rather than replacing them.
- A dedicated appliance creates a new hardware category without requiring users to buy a traditional computer.

10. The Hard Questions

Why not just use the iPhone? Because the largest models will continue to demand more memory and sustained compute than a thin, battery-powered device can economically provide.

Why not just use the cloud? Local compute can reduce latency, improve resilience and keep more personal context inside the home.

Why would consumers buy another device? The value proposition is not the box itself; it is making every Apple device in the household substantially more capable.

What about electricity? The device should be designed around efficient inference and low idle power, with aggressive scheduling and model selection.

Would this replace Macs? No. **It is an inference appliance, not a general-purpose computer. A Mac remains the user's computer; Apple Neural is the household's AI compute layer.**

Is the market big enough? The strongest path may be to position it initially for Apple Intelligence power users, then expand its role as AI agents and smart-home capabilities mature.

11. A Possible Apple Roadmap

Phase	Theme	What it unlocks
Phase 1	Home inference	Larger Siri and Apple Intelligence models locally.
Phase 2	Personal context	Secure local indexes across permitted photos, files, calendars and preferences.

Phase 3	Home agent	Natural-language HomeKit orchestration and proactive routines.
Phase 4	Agentic compute	Long-running personal workflows that can continue even when the user's phone is unavailable.
Phase 5	Compute fabric	Optional coordination across Macs and other Apple Silicon devices in the home.

12. Why This Is Worth Discussing Now

The strategic question is shifting from “How do we put AI on every device?” to “Where should the intelligence live?” A household AI compute layer offers a middle ground: more capable than edge devices, more private than a traditional cloud, and more economical than putting maximum compute into every product.

ONE-SENTENCE PITCH

Apple should build a Mac-mini-sized AI appliance that becomes the private intelligence center for every Apple device in your home.

Suggested Twitter/X launch thread

- 1. I have a product idea for Apple: Apple Neural — a dedicated Apple Silicon AI computer for your entire home.**
- 2.** The problem: the best AI models are getting too large for phones and watches, but sending everything to the cloud has latency and privacy trade-offs.
- 3.** What if Apple sold a small, silent compute box that runs larger Apple Intelligence models locally?
- 4. Your iPhone asks a complex question → sends it over your home Wi-Fi → Apple Neural runs the model → response comes back in milliseconds.**
- 5.** One device. Multiple family members. Separate private contexts. Shared compute.
- 6. Simple requests stay on-device. Bigger requests go to Apple Neural. Only the largest workloads escalate to Apple's private cloud.**
- 7.** This isn't another Mac. It's an AI appliance: the compute layer between your devices and the cloud.

8. The interesting future: your Mac, iPhone, iPad and Apple Neural could become a small private AI compute fabric.

9. Apple already owns the silicon, OS, Siri, Apple Intelligence, HomeKit, identity and private cloud pieces. The missing layer may be the box in the middle.

10. Would you buy one?

13. Closing Thought

The long-term opportunity is bigger than local inference.

If the personal computer becomes an AI agent, then the home may need its own compute layer.

Apple is uniquely positioned to make that layer invisible: a small box that quietly turns every Apple device around you into a more capable computer.

APPLE NEURAL

Private AI. Shared compute. One Apple home.